

Cours MVSI
Modélisation et Vérification
des Systèmes Informatiques

Sommaire du cours

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- ① Tracking bugs in C codes
- ② Introduction by Example
 - Detecting overflows in computations
 - Computing the velocity of an aircraft on the ground
- ③ Verification of program properties
- ④ Topics of course

Listing 1 – Bug bug0

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int x, y;
    // Seed the random number generator with the current time
    srand(time(NULL));
    // Generate a random number between 1 and 100
    x = rand() % 100 + 1;
    // Perform some calculations
    y = x / (100 - x);
    printf("x=%d--and-y=%d\n", x, y);
    return 0;
}
```

Listing 2 – Bug bug0

```
#include <stdio.h>
#include <stdlib.h>
#include <time.h>

int main() {
    int x, y;
    // Seed the random number generator with the current time
    srand(time(NULL));
    // Generate a random number between 1 and 100
    x = rand() % 100 + 1;
    // Perform some calculations
    y = x / (100 - x);
    printf("x=%d and y=%d\n", x, y);
    return 0;
}
```

bug0.c prints x =some number and y =some other number

Listing 3 – Bug bug00

```
// Heisenbug
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
//
int main() {
    int x, y, i=0;
    for (i = 0; i <= 1000000; i++) {
        // Seed the random number generator with the current time
        srand(time(NULL));
        // Generate a random number between 1 and 100
        x = rand() % 100 + 1;
        printf("Result: -x=-%d\n", x);
        // Perform some calculations
        y = x / (100 - x);
        printf("Result: -i=%d--and-y=%d\n", i, y);
    }
    return 0;
}
```

Listing 4 – Bug bug00

```
// Heisenbug
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
//
int main() {
    int x, y, i=0;
    for (i = 0; i <= 1000000; i++) {
        // Seed the random number generator with the current time
        srand(time(NULL));
        // Generate a random number between 1 and 100
        x = rand() % 100 + 1;
        printf("Result: -x=-%d\n", x);
        // Perform some calculations
        y = x / (100 - x);
        printf("Result: -i=%d--and-y=%d\n", i, y);
    }
    return 0;
}
```

**bug00.c prints ... i=some
number and y=some other number
...**

Listing 5 – Bug bug000

```
// Heisenbug
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <limits.h>

int main() {
    int x, y, i=0;

    for (i = 0; i <= INT_MAX; i++) {
        // Seed the random number generator with the current time
        srand(time(NULL)+i);

        // Generate a random number between 1 and 100
        x = rand() % 100 + 1;

        printf("Result: -x=-%d\n", x);
        // Perform some calculations
        y = x / (100 - x);

        printf("Result: -i=%d -x=-%d -and -y=%d\n", i, x, y);
    }

    return 0;
}
```

Listing 6 – Bug bug000

```
// Heisenbug
#include <stdio.h>
#include <stdlib.h>
#include <time.h>
#include <limits.h>

int main() {
    int x, y, i=0;

    for (i = 0; i <= INT_MAX; i++) {
        // Seed the random number generator with the current time
        srand(time(NULL)+i);

        // Generate a random number between 1 and 100
        x = rand() % 100 + 1;

        printf("Result: -x=-%d\n", x);
        // Perform some calculations
        y = x / (100 - x);

        printf("Result: -i=%d -x=-%d -and -y=%d\n", i, x, y);
    }

    return 0;
}
```

Result: x= ??

Result: i=?? x=?? and y=??

- ▶ RTE means Run Time Error
- ▶ $x = \text{rand}() \% 100 + 1$ and $y = x / (100 - x)$
- ▶ $\text{rte}(x = \text{rand}() \% 100 + 1) \stackrel{\text{def}}{=} \text{rte}(x) \wedge \text{rte}(\text{rand}()) \wedge \text{rte}(\text{rand}() \% 100 + 1)$
- ▶ $\text{rte}(y = x / (100 - x)) \stackrel{\text{def}}{=} \text{rte}(y) \wedge \text{rte}(x) \wedge 100 \neq x$

RTE

The value of x may be 100 and a sequence of test can not get x with the value 100 .

Listing 7 – Bug bug1

```
#include <stdio.h>
#define n 5
int main() {
    int numbers[n] = {1, 2, 3, 4, 5};
    int sum = 0;

    // Attempt to calculate the sum of numbers in the array
    for (int i = 0; i <= n; i++) {
        sum += numbers[i];
    };
    int val = n*(n+1)/2;
    printf(" Iteratively -computed -sum=%d -and -direct -comutation -val=%d\n" , sum, val);
    return 0;
}
```

Listing 8 – Bug bug1

```
#include <stdio.h>
#define n 5
int main() {
    int numbers[n] = {1, 2, 3, 4, 5};
    int sum = 0;

    // Attempt to calculate the sum of numbers in the array
    for (int i = 0; i <= n; i++) {
        sum += numbers[i];
    };
    int val = n*(n+1)/2;
    printf(" Iteratively -computed- sum=%d -and- direct -comutation- val=%d\n", sum, val);
    return 0;
}
```

Iteratively computed sum=16 and
direct comutation val=15

Listing 9 – Bug bug2

```
#include <stdio.h>

int main() {
    int x = 5;
    int y = 3;

    // Bug 1: Incorrect variable in the printf statement
    printf("The value of x is: %d\n", y);

    // Bug 2: Infinite loop
    while (x > 0) {
        printf("x is greater than 0\n");
    }

    return 0;
}
```

Listing 10 – Bug bug2

```
#include <stdio.h>

int main() {
    int x = 5;
    int y = 3;

    // Bug 1: Incorrect variable in the printf statement
    printf("The-value-of-x-is:-%d\n", y);

    // Bug 2: Infinite loop
    while (x > 0) {
        printf("x-is-greater-than-0\n");
    }

    return 0;
}
```

Infinite loop . . .

Listing 11 – Bug bug7

```
#include <stdio.h>
#include <limits.h>
int average(int a, int b)
{
    return ((a+b)/2);
}

int main()
{
    int x, y;
    x=INT_MAX; y=INT_MAX;
    printf(" Average -- for -%d- and -%d- is -%d\n" , x, y,
        average(x, y));
    return 0;
}
```

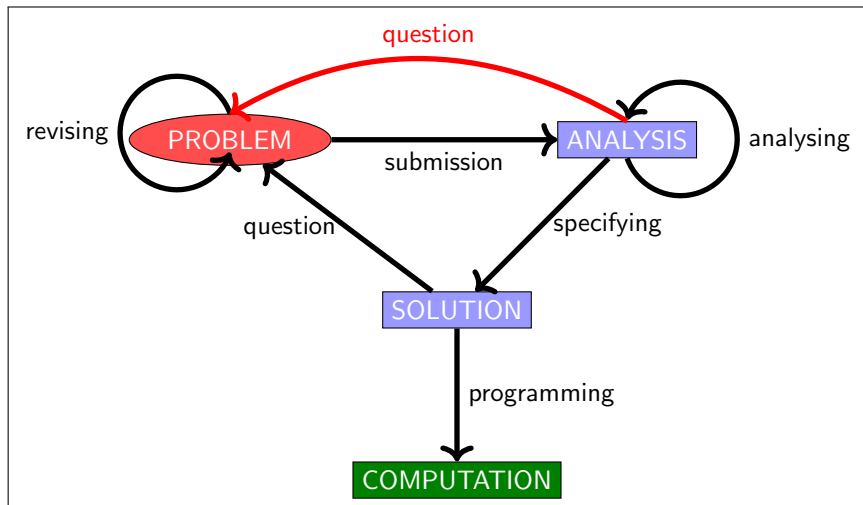
Listing 12 – Bug bug7

```
#include <stdio.h>
#include <limits.h>
int average(int a, int b)
{
    return ((a+b)/2);
}

int main()
{
    int x, y;
    x=INT_MAX; y=INT_MAX;
    printf(" Average -- for %d - and %d - is %d\n", x, y,
        average(x, y));
    return 0;
}
```

Average for 2147483647 and
2147483647 is -1

Problem versus Solution



Listing 13 – Function average

```
#include <stdio.h>
#include <limits.h>
int average(int a,int b)
{
    return((a+b)/2);
}

int main()
{
    int x,y;
    x=INT_MAX;y=INT_MAX;
    printf(" Average - - for -%d- and -%d- is -%d\n" ,x,y ,
           average(x,y));
    return 0;
}
```

Execution produces a result

Average for 2147483647 and 2147483647 is -1

Execution produces a result

Average for 2147483647 and 2147483647 is -1

Using frama-c produces a required annotation

```
int average(int a, int b)
{
    int __retres;
    /*@ assert rte: signed_overflow: -2147483648 <= a + b; */
    /*@ assert rte: signed_overflow: a + b <= 2147483647; */
    __retres = (a + b) / 2;
    return __retres;
}
```

Listing 14 – Function average.....

```
#include <stdio.h>
#include <limits.h>
/*@ requires 0 <= a;
    requires a <= INT_MAX ;
    requires 0 <= b;
    requires b <= INT_MAX ;
    requires 0 <= a+b;
    requires a+b <= INT_MAX ;
    ensures \result <= INT_MAX;

*/
int average(int a,int b)
{
    return((a+b)/2);
}

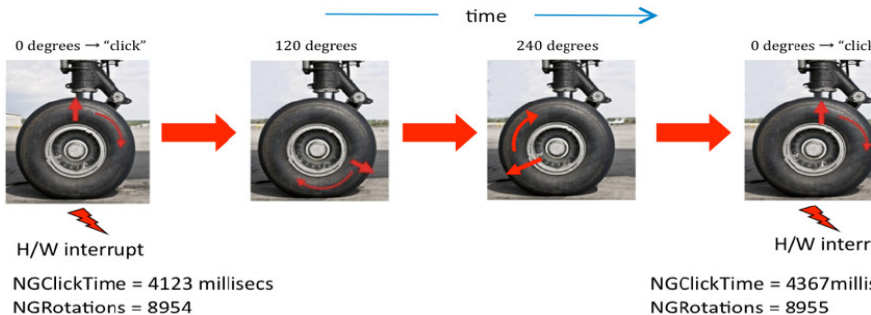
int main()
{
    int x,y;
    x=INT_MAX / 2;y=INT_MAX / 2;
    // printf("Average for %d and %d is %d\n",x,y,
    // );
    return average(x,y);
}
```



- ▶ Estimated ground velocity of the aircraft should be available only if it is within 3 km/hr of the true velocity at some moment within past 3 seconds

Characterization of a System (I)

- ▶ **NG velocity system :**
 - **Hardware :**
 - ▶ *Electro-mechanical sensor* : detects rotations
 - ▶ *Two 16-bit counters* : Rotation counter, Milliseconds counter
 - ▶ *Interrupt service routine* : updates rotation counter and stores current time.
 - **Software :**
 - ▶ *Real-time operating system* : invokes update function every 500 ms
 - ▶ *16-bit global variable* : for recording rotation counter update time
 - ▶ *An update function* : estimates ground velocity of the aircraft.
- ▶ **Input data available to the system :**
 - *time* : in milliseconds
 - *distance* : in inches
 - *rotation angle* : in degrees
- ▶ **Specified system performs velocity estimations in *imperial* unit system**
- ▶ **Note** : expressed functional requirement is in *SI* unit system (km/hr).



WHEEL_DIAMETER = 22 inches
PI = 3.14

12 inches/foot
5280 feet/mile

$$\begin{aligned}\text{estimatedGroundVelocity} &= \text{distance travel/elapsed time} \\ &= ((3.14 * 22)/(12*5280))/((4367-4123)/(1000*3600)) \\ &= 16 \text{ mph}\end{aligned}$$

Safety Property

- ▶ Storing the number of *NGClick* in a *n*-bit variable *VNGClick*
- ▶ Integers are denoted by the set *Int* and is simply defined by the interval $Int \hat{=} INT_MIN..INT_MAX$.
- ▶ Safety requirement :
The value of VNGClick is always in the range of implementation Int or equivalently $VNGClick \in Int$
- ▶ $Length = \pi \cdot diameter \cdot VNGClick$ (mathematical property)
- ▶ $Length \leq 6000$ (domain property)
- ▶ $\pi \cdot diameter \cdot VNGClick \leq 6000$
- ▶ $VNGClick \leq 6000 / (\pi \cdot diameter)$
- ▶ if $n=8$, then $2^7 - 1 = 127$ and
 $6000 / (\pi \cdot [22, inch]) = 6000 / (\pi \cdot 55, 88) = 6000 / (3, 24 \cdot [55, 88, cm]) = 6000 / (3, 24 \cdot 0.5588) \approx 3419$ and the condition of safety can not be satisfied in any situation.
- ▶ if $n=16$, then $2^{15} - 1 = 65535$ and $6000 / (\pi \cdot [22, inch]) \approx 3419$ and the condition of safety can be satisfied in any situation since

Safety Property

- ▶ Storing the number of `NGClick` in a `n`-bit variable `VNGClick`
- ▶ Integers are denoted by the set *Int* and is simply defined by the interval $Int \hat{=} INT_MIN..INT_MAX$.
- ▶ Safety requirement :
*The value of `VNGClick` is always in the range of implementation *Int* or equivalently $VNGClick \in Int$*

$$RTE_VNGClick : 0 \leq vNGClick \leq INT_MAX \quad (1)$$

- ▶ The current value of `VNGClick` is always bounded by the two values 0 and `INT_MAX`.

- ▶ Typing Properties using Typechecker (see for instance functional programming languages as ML, CAML, OCAML, ...)
- ▶ Invariance and safety (*A nothing bad will happen !*) properties for a program P :
 - Transformation of P into a relational model M simulating P
 - Expression of safety properties :
$$\forall s, s' \in \Sigma. (s \in \text{Init}_S \wedge s \xrightarrow{*} s') \Rightarrow (s' \in A).$$
 - Definition of the set of reachable states of P using M :
$$\text{REACHABLE}(M) = \text{Init}_S \cup \longrightarrow [\text{REACHABLE}(M)]$$
 - Main property of $\text{REACHABLE}(M)$: $\text{REACHABLE}(M) \subseteq A$
 - Characterization of $\text{REACHABLE}(M)$:
$$\text{REACHABLE}(M) = \text{FP}(\text{REACHABLE}(M))$$

- ▶ Proving automatically $\text{REACHABLE}(M) \subseteq A$:

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- ▶ Proving automatically $\text{REACHABLE}(M) \subseteq A$: changing the domain and solving in another domain as abstract interpretation if making possible
- ▶ Proving automatically $\text{REACHABLE}(M) \subseteq A$: approximating semantics of programs

- A problem $x \in P$ is generally stated by the function $\chi_{x \in P}$ where $\chi_{x \in P}(u) = 1$, if $P(u)$ is true and $\chi_{x \in P}(u) = 0$, if $P(u)$ is false :
- Problem 1 : $x \in 0..n$ where $n \in \mathbb{N}$
 - Problem 1 : $w \in \mathcal{L}(G)$ where G is a grammar over the finite set of alphabet symbols Σ and $\mathcal{L}(G) \subseteq \Sigma^*$.

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- ▶ A problem $x \in P$ is decidable, when the function $\chi_{x \in P}$ is computable or more precisely the function can be computed by a program
- ▶ Problem of the correctness of a program :
 - Assume that $\mathcal{F}$ is the set of unary function over natural numbers : $\mathcal{F} = \mathbb{N} \rightarrow \mathbb{N}$.
 - $\mathcal{C} \subseteq \mathcal{F}$: the set of computable (or programmable) functions is $\mathcal{C}$
 - $f \in \mathcal{C} = \{\Phi_0, \Phi_1, \dots, \Phi_n, \dots\}$: the set of computable functions is denumerable.
 - The problem $x \in \text{dom}(\Phi_y)$ is not decidable and it expresses the correctness of programs.

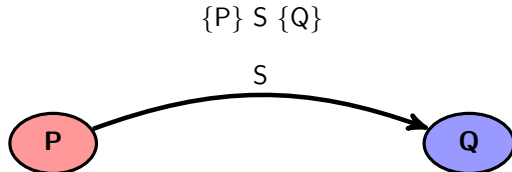
Implicite versus explicite

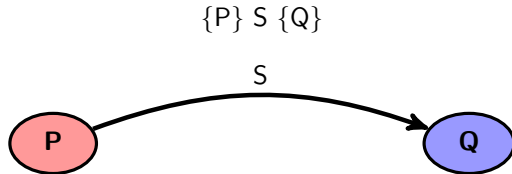
- Ecrire $101 = 5$ peut avoir une signification

Example : description of static behaviour

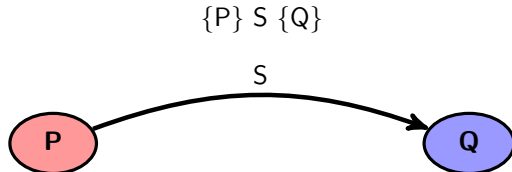
- ▶ A train moving at absolute speed *spd1*
- ▶ A person walking in this train with relative speed *spd2*
 - One may compute the absolute speed of the person
- ▶ Modelling
 - Syntax. Classical expressions
 - ▶ Type *Speed* = *Float*
 - ▶ *spd1, spd2* : *Speed*
 - ▶ *AbsoluteSpeed* = *spd1* + *spd2*
 - Semantics
 - ▶ If *spd1* = 25.6 and *spd2* = 24.4 then *AbsoluteSpeed* = 50.0
 - ▶ If *spd1* = "val" and *spd2* = 24.4 then exception raised
 - Pragmatics
 - ▶ What if *spd1* is given in *mph* (miles per hour) and *spd2* in *km/s* (kilometers per second) ?
 - ▶ What if *spd1* is a relative speed ?

- ▶ Un programme P *produit* des résultats à partir de données en accord avec une sémantique :
 - STATES est l'ensemble de tous les états de P : $STATES = x \rightarrow \mathbb{Z}$ où x désigne les variables de P.
 - s_0 et s_f deux états de STATES : $\mathcal{D}(P)(s_0) = s_f$ signifie que P est exécuté à partir d'un état s_0 et produit un état s_f .
 - Pour un état s de P courant, on notera $s(x) = x$ pour distinguer la valeur de la variable x et sa valeur courante en s :

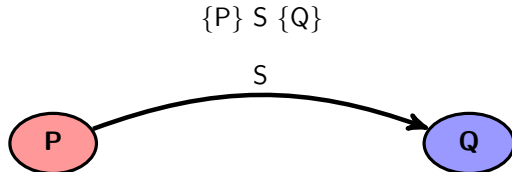




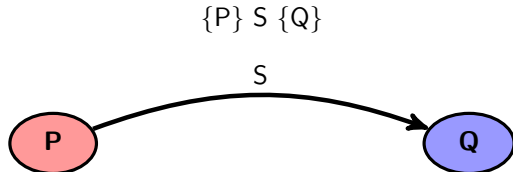
► $\{P\} S \{Q\}$: *asserted program*



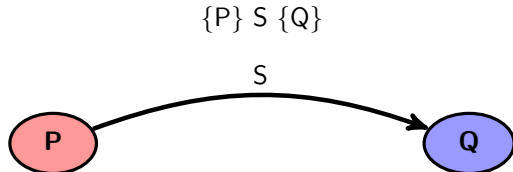
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- ▶ $SP(S)(P) \Rightarrow Q$: *logical formula*



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- ▶ $SP(S)(P) \Rightarrow Q$: *logical formula*
- ▶ $C(S) \vdash SP(S)(P) \Rightarrow Q$: *logical formula*

Predicate Transformer

$WP(S)(Q)$ is the Weakest-Precondition of S for Q and is a predicate transformer but $WP(S)(.)$ is not a computable function over the set of predicates.

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- ▶ Outils

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 - 😊 Vérification de propriétés de systèmes avec un analyseur Frama-c
- ▶ Outils
 - 😊 TLA/TLA⁺ avec TLC, Frama-c

- ▶ Découpage de l'unité : 10 cours de 2 h
- ▶ Contenu :
 - 😊 Principes de modélisation des systèmes informatiques : systèmes de transition
 - 😊 Propriétés d'un système informatique : sûreté, vivacité, disponibilité, sécurité, dépendabilité
 - 😊 Modélisation de propriétés de systèmes
 - 😊 Analyse des programmes
 - 😊 Vérification de propriétés de systèmes avec un model checker TLC
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